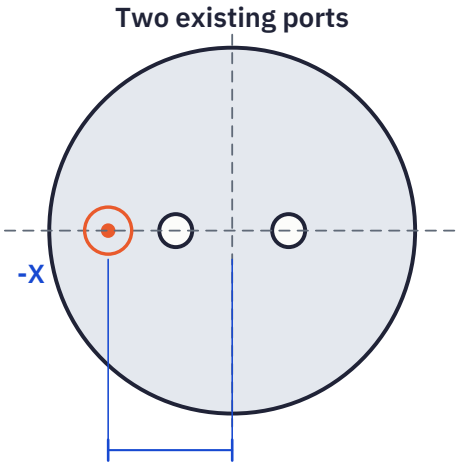
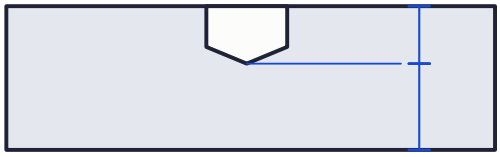
DRILL THE BLIND REGISTER

One small pocket, inside each endcap.
It locates the float rod without piercing the plate.

INSIDE FACE / TOP VIEW



ENLARGED SECTION



9/64 in drill

0.100 in to TIP

0.150 in stays solid

Do not drill through.

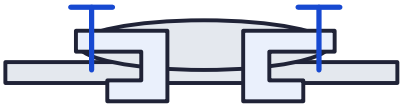
Ports' midpoint = plate center. The register lies beyond the negative port on their diameter.

1 Locate the pocket



Inside face up. Mark the location shown above on both plates.

2 Clamp the disc



Secure it to the press table. Check the bit and chuck clear every clamp.

3 Set + prove the stop



Touch the stopped tip to the face. Set 0.100 in travel; prove on scrap.

4 Drill both plates



9/64 in M35 cobalt, 135° split point. ~740 RPM; Tap Magic on the point.

CHECK BOTH POCKETS

Blind, 0.100 in to tip; 0.150 in of solid plate remains.

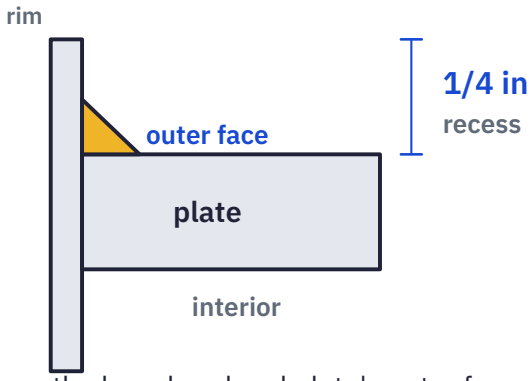
Sources: pressure-vessel §1 / endcap drawing. Dimensions govern; picture is not a template.

SET UP THE CLOSURE WELD

Working end up. Endcap's outer face toward you.
The register face and float rod face into the tube.

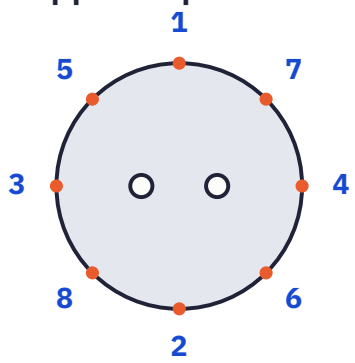
Class 4 laser. Laser-rated eye/face protection, gloves, protective clothing.
Use a protected welding area with fume extraction.

1 Clean + seat



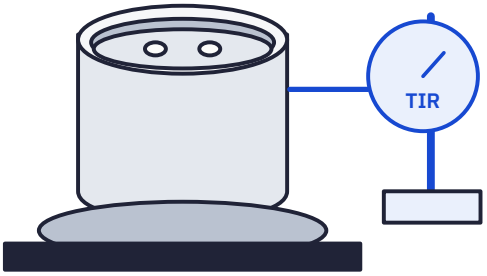
Clean the bore band and plate's outer face.
Deburr tube ID + OD.
Seat on the 1/4 in spacer; check all around.

2 Tack opposite pairs



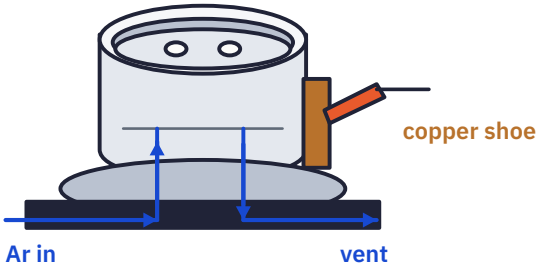
Eight tacks, bisecting the gaps in opposite pairs.
Recheck seated depth.
The rod must not hold the plate up.

3 Indicate the work



Clamp the base; touch adjusters lightly.
Weld-end radial TIR <= 0.25 mm.
Endcap-face TIR <= 0.30 mm.
Use the commissioned rotator.

4 Purge + prove contact



First: purge through open held end.
Second: lower port in; other port vent.
Scuff tube + shoe; clip work lead to shoe.
Laser OFF: continuity for 2 dry turns.

POWER	WOBBLE	WIRE	ARGON	FILLER
60%	80 Hz / 2 mm	12 mm/s	2 s pre/post	ER316L .030

Recorded practice recipe. Closure root fusion and internal oxidation still need qualification. These settings are for the endcap closure.

Sources: pressure-vessel §§2-5 / weld-rotation-rig / X1 Pro manual, pp.11-14.

START THE BEAD

Your foot turns the tube. Your trigger starts the weld.
Hold the head steady as the joint travels under it.

1 Set speed while stopped



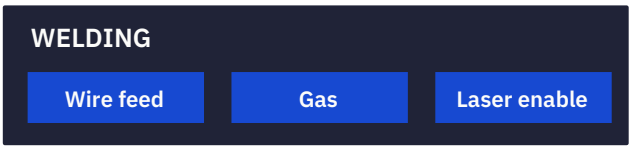
Start at 8 mm/s travel. Verify direction from above and return the index to the first tack.

2 Aim into the corner



Beam into plate mass, washing onto wall. Wire on arriving side. Use the coupon-proven angle + standoff.

3 Ready the welder



PPE + protected area ready; shielding and back-purge established. Enable wire feed and laser on the X1 Pro.

4 Press + hold the pedal



Keep the trigger released. Hold the pedal down until rotation is steady; keep the head at the joint.

KNOW THE FINISH BEFORE YOU START

Carry the bead a full lap, then about 20° past the first tack. Trail off; **release trigger, then pedal.** Keep the head in position for post-flow. The table has no automatic stop. Snip stuck wire with the gun held still.

5 Pedal held. Pull + hold the trigger.

